

GNU Coreutils Compatibility Audit

Typed C-subset compiler boundary

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Abstract

The upstream GNU coreutils source has been downloaded and inspected as a compatibility target for the strongly typed compiler. This audit identifies the real dependencies of standard tools and defines an incremental path toward compiling them without pretending that the current small language already supports full C.

1 Downloaded source

The checkout is in `coreutils-src/`, revision `d458e11`. Representative files are 73 lines for `true.c`, 286 for `echo.c`, and 386 for `pwd.c`. The source is real GNU coreutils, not toy replacements.

2 Observed dependencies

Even `true.c` includes generated `config.h`, project `system.h`, and `true.h`; accepts `argc/argv`; calls initialization, localization, diagnostics, and exit helpers; and uses pointers and external `libc/project` APIs. `echo.c` adds pointer arithmetic, option processing, character classification, variadic I/O, and project macros. `pwd.c` adds structs, allocation, filesystem APIs, and long-option parsing.

The current compiler intentionally supports a much smaller typed subset. It cannot honestly compile these files directly yet, even if macro expansion is delegated to a preprocessor. Macro removal does not remove the required C type and runtime surface.

3 Compatibility boundary

The principled pipeline is:

C source $\xrightarrow{\text{preprocess}}$ C AST $\xrightarrow{\text{typed translation}}$ typed IR $\xrightarrow{\text{normalize}}$ residual IR $\xrightarrow{\text{ABI lowering}}$ machine code.

The next required IR features are pointers, arrays, structs, function calls, and explicit `libc/syscall` effects. The linker surface must be declared rather than implicitly inherited.

4 Milestones

The first practical milestone is a dependency-reduced freestanding version of ‘true’, followed by a small ‘echo’ slice. Only after those pass should the project attempt ‘pwd’, ‘ls’, or ‘find’, whose filesystem and option APIs are substantially larger.

5 Conclusion

The source set is downloaded and inventoried. Full GNU compatibility is not yet achieved; the audit identifies the exact missing language and runtime boundaries required to proceed soundly.